

FEB 2023 EXAMINATION
I B.E./B.TECH. (4YDC) EXAM
EE 10015/10005: FUNDAMENTALS OF ELECTRICAL ENGINEERING

Time: 3 Hrs.

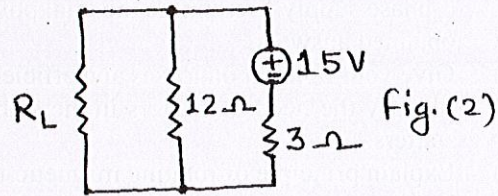
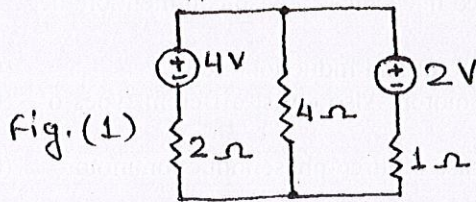
[Max. Marks: 70]

[Min. Pass Marks: 22]

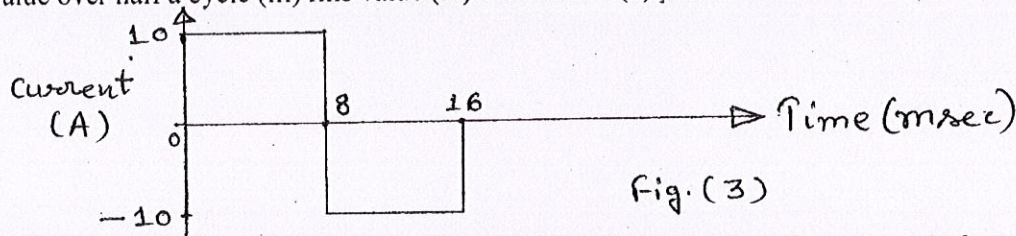
TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

Note: Attempt all the FIVE questions. Internal choice is there, see carefully and attempt accordingly.

- | | | | Marks | COs | BL |
|-----|---|---|-------|-----|----|
| Q.1 | (a) | Write the statement of - Ohm's law, Kirchhoff's Voltage law and Kirchhoff's current law. | (03) | 1 | 1 |
| | (b) | Prepare the list of steps which are used in mesh analysis and nodal analysis for the solution of electrical circuit. | (04) | 1 | 1 |
| | (c) | Figure (1) shows a circuit containing two sources of emf, each with their internal resistance. Determine the current in each branch of the network by using the superposition theorem. Also write the statement of superposition theorem. | (07) | 1 | 5 |
| | OR | | | | |
| (d) | Find the value of the load resistor R_L shown in figure (2) that gives maximum power dissipation and determine the value of this power. Also write the statement of maximum power transfer theorem. | (07) | 1 | 5 | |



- Q.2 (a) For the periodic waveforms shown in figure (3), determine (i) frequency (ii) average value over half a cycle (iii) rms value (iv) form factor (v) peak factor. (03) 2 5



- | | | | | | |
|--|-----------|--|------|---|---|
| | (b) | Draw the circuit diagram, phasor diagram, voltage triangle and impedance triangle for – (i) RL series single phase AC circuit (ii) RC series single phase AC circuit. | (04) | 2 | 1 |
| | (c) | A coil of inductance 159.2mH and resistance 40 ohm is connected in parallel with a 30μF capacitor across a 230V, 50Hz supply. Determine – (i) the current in the coil and its phase angle (ii) the current in the capacitor and its phase angle (iii) the supply current and its phase angle (iv) the circuit impedance (v) the power consumed (vi) the apparent power and (vii) the reactive power. Draw the phasor diagram. | (07) | 2 | 5 |
| | OR | | | | |
| | (d) | A motor has an output of 4.8kW, an efficiency of 80% and a power factor of 0.625 lagging when operated from a 230V, 50Hz supply. It is required to improve the power factor to 0.95 lagging by connecting a capacitor in parallel with the motor. Determine – (i) the current taken by the motor (ii) the supply current after power factor correction (iii) the current taken by the capacitor (iv) the capacitance of the capacitor, and (v) the kVAR rating of the capacitor. | (07) | 2 | 5 |

P.T.O.

- Q.3 (a) Prove that in the case of transformer $(V_1/V_2) = (N_1/N_2) = (I_2/I_1)$. (03) 3 5
 (b) Draw and explain hysteresis curve for ferromagnetic material with proper labelling. (04) 3 2
 (c) A closed magnetic circuit of cast steel contains a 6 cm long path of cross-sectional area 1 cm^2 and a 2 cm path of cross-sectional area 0.5 cm^2 . A coil of 200 turns is wound around the 6 cm length of the circuit and a current of 0.4 A flows. Determine the flux density in the 2 cm path, if the relative permeability of the cast steel is 750. Also draw the magnetic circuit and electrical equivalent circuit for the given problem. (07) 3 5

OR

- (d) A 400kVA transformer has a primary winding resistance of 0.5 ohm and a secondary winding resistance of 0.001 ohm. The iron loss is 2.5kW and the primary and secondary voltages are 5kV and 320V respectively. If the power factor of the load is 0.85, determine the efficiency of the transformer (i) on full load, and (ii) on half load. Also draw the approximate equivalent circuit of the transformer. (07) 3 5

- Q.4 (a) Define – Prime mover, Synchronous generator and Line voltage. (03) 4 1
 (b) Give the reason, why two-wattmeter method is more popular than three-wattmeter method. Also draw circuit diagram of both the methods. (04) 4 1
 (c) Three identical coils, each of resistance 10 ohm and inductance 50 mH are connected (i) in star and (ii) in delta to a 415V, 50Hz, 3-phase supply. Determine the total power dissipated in each case. Also comment on the obtained answers. (07) 4 5

OR

- (d) The three arms of a 3-phase load each comprise of resistance of 15 ohm and capacitor in series with a value of $150 \mu\text{F}$ are connected (i) in star and (ii) in delta to a 415V, 50Hz, 3-phase supply. Determine the total power dissipated in each case. Also comment on the obtained answers. (07) 4 5

- Q.5 (a) Give your opinion on losses and efficiency of DC motor and Induction motor. (03) 5 5
 (b) Identify the need of starters in the starting of DC motor. Also enlist different types of starters. (04) 5 3
 (c) Explain principle of rotating magnetic field in the case of three-phase induction motor. (07) 5 2

OR

- (d) Illustrate the working principle of DC shunt motor. (07) 5 2

JULY
JUNE 2022 EXAMINATION
I B.E./B.TECH (4YDC) EXAM

EE 10005: FUNDAMENTALS OF ELECTRICAL ENGINEERING

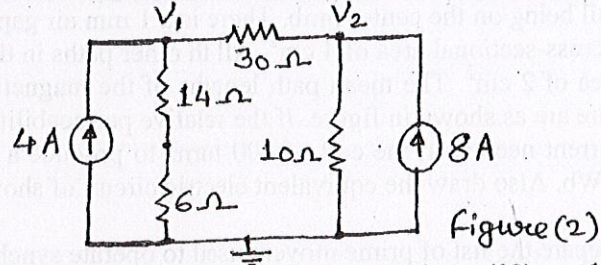
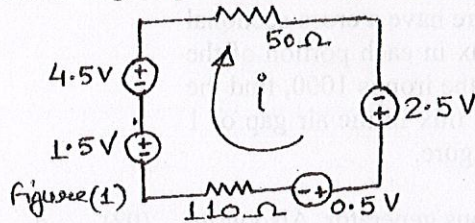
Time: 3 Hrs.

756
[Max. Marks: 70]
[Min. Pass Marks: 22]

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

Note: Attempt all the FIVE questions. Internal choice is there, please see carefully and attempt accordingly. CO – Course Outcome, BL – Bloom's Taxonomy Level.

- | | | Marks | CO | BL |
|-----|--|-------|----|----|
| Q.1 | (a) Determine the current i in the circuit of figure (1) by first combining the sources into a single equivalent voltage source. | (02) | 1 | 3 |



- | | | | | |
|--|---|------|---|-----|
| | (b) State the effect of an increase in temperature on the resistance of each of the following: aluminium, copper, tungsten, and carbon. | (02) | 1 | 2 |
| | (c) Determine the current flowing left to right through the 30Ω resistor of figure (2), using nodal analysis. | (03) | 1 | 3 |
| | (d) Determine the current through 4Ω resistor in the network shown in figure (3), by the use of Thevenin's theorem. Also write statement of the used theorem. | (07) | 1 | 1,3 |

OR

- | | | | | |
|--|---|------|---|-----|
| | (e) Determine the current through 4Ω resistor in the network shown in figure (3), by the use of Norton's theorem. Also write statement of the used theorem. | (07) | 1 | 1,3 |
|--|---|------|---|-----|

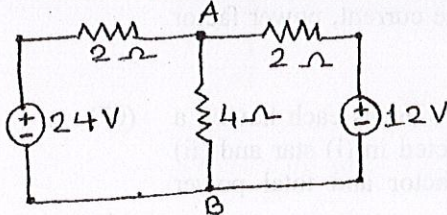


Figure (3)

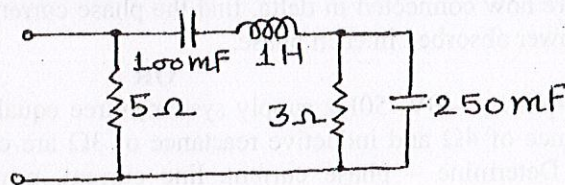


Figure (4)

- | | | | | |
|-----|--|------|---|---|
| Q.2 | (a) Write the specification of 1-φ AC supply used in the household applications. Also write at least two effects of poor power factor on the connected system. | (02) | 2 | 1 |
| | (b) Define phasor diagram. Enlist limitations of phasor diagram. | (02) | 2 | 1 |
| | (c) Determine the equivalent impedance of the network shown in figure (4), given an operating frequency of 5 rad/sec. | (03) | 2 | 3 |
| | (d) A circuit consists of two coils in series connected to a 200V AC supply. The first coil has a resistor of 10Ω and inductive reactance of 20Ω. The second coil has a resistance of 12Ω and inductive reactance of 16Ω. Find – the total impedance of the circuit, the current in the circuit, the circuit phase angle, the voltage drop in each coil, total power consumed by the circuit, power consumed by the first coil as well as second coil. | (07) | 2 | 3 |

OR

- | | | | | |
|--|---|------|---|---|
| | (e) Three impedances $(6 + j 5) \Omega$, $(8 - j 6) \Omega$, and $(8 + j 10) \Omega$ are connected in parallel. Calculate the current in each branch when the total current is 20A. Also calculate power consumed by the individual branch and total power consumed by the circuit. | (07) | 2 | 3 |
|--|---|------|---|---|

P.T.O.

- Q.3 (a) What is meant by leakage flux and fringing in association with magnetic circuit? (02) 3 1
 (b) Derive the EMF equation of a transformer. (02) 3 2
 (c) Define the terms – Hysteresis loss and Eddy current loss. Also write their mathematical expression. (03) 3 1
 (d) A transformer is rated at 150kVA. At full load its copper loss is 1250W and its iron loss is 1000W. Find – (07) 3 3
 (i) the efficiency at full load, unity power factor
 (ii) the efficiency at half load, 0.8 power factor
 (iii) the efficiency at 75% load, 0.7 power factor
 (iv) the load kVA at which maximum efficiency will occur
 (v) the maximum efficiency at 0.85 power factor.

OR

- (e) An inductor has a core built up of stampings of the shape shown in figure (5), the coil being on the center limb. There is a 1 mm air gap in the center limb which has a cross-sectional area of 4 cm^2 . All the other paths in the core have a cross-sectional area of 2 cm^2 . The mean path lengths of the magnetic flux in each portion of the core are as shown in figure. If the relative permeability of the iron is 1000, find the current needed in the coil of 600 turns to produce a total flux in the air gap of 1 mWb. Also draw the equivalent electric circuit of shown figure. (07) 3 3

- Q.4 (a) Prepare the list of prime movers used to operate synchronous generator. Also write function of diesel – generator set. (02) 4 1
 (b) Draw the connection diagram of a 3-phase, four wire supply system with proper labeling. (02) 4 1
 (c) Two wattmeter are used to measure the power in a 3-phase balanced system. What is the power factor when (i) both the meters read equal (ii) both the meters read equal but one is negative (iii) one reads twice the other? (03) 4 2
 (d) Three similar coils each having a resistance of 30Ω and an inductance of 0.05H are connected in star to a 3-phase, 50Hz supply, 230 volts between the lines. Find the phase current, line current, power factor and the power absorbed in each phase. If they are now connected in delta, find the phase current, line current, power factor and power absorbed in each phase. (07) 4 3

OR

- (e) In a 3-phase, 440V, 50Hz supply system, three equal impedances each having a resistance of 4Ω and inductive reactance of 3Ω are connected in (i) star and (ii) delta. Determine – phase current, line current, power factor and total power consumed. (07) 4 3

- Q.5 (a) Write at least four applications of DC motor. Also write speed relation of DC motor. (02) 5 1
 (b) What are the different losses occur in an induction motor, show with the help of power flow diagram. (02) 5 1
 (c) Demonstrate the working of 1-phase fan motor. (03) 5 2
 (d) Prepare the technical note on 3-phase induction motor. (07) 5 2

OR

- (e) Prepare the technical note on DC motor. (07) 5 2

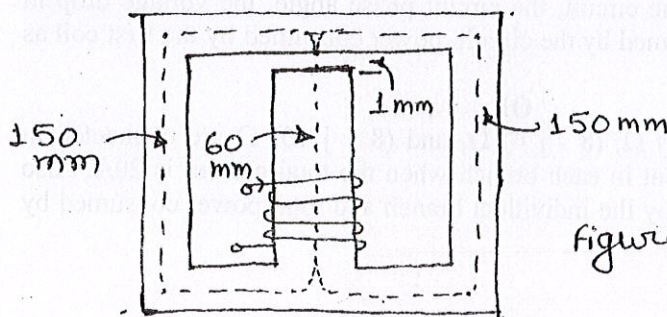


Figure (5)

ONLINE FEBRUARY 2022 EXAMINATION

I B.E./B.TECH (4YDC) EXAM

EE 10005: FUNDAMENTALS OF ELECTRICAL ENGINEERING

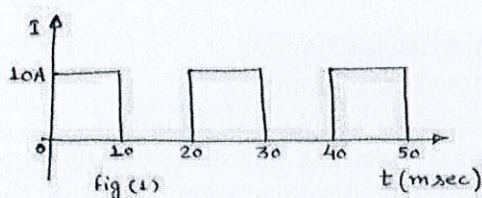
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[Max. Marks: 40

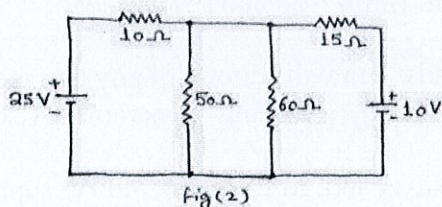
TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

Note: Attempt all the FIVE questions. Internal choice is there, please see carefully and attempt accordingly. CO – Course outcome, BL – Bloom's Taxonomy Level

- | | Marks | CO | BL |
|--|-------|----|-----|
| Q.1 (a) A non-alternating periodic waveform has been shown in fig. (1). Calculate its form factor. | (02) | 1 | BL1 |

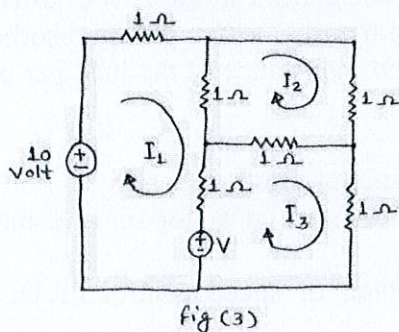


- | | | | |
|--|------|---|-----|
| (b) Write the statement of superposition and Norton's theorem. | (02) | 1 | BL1 |
| (c) Find the current through 50Ω resistor shown in fig. (2), using nodal analysis method. | (04) | 1 | BL2 |



OR

- | | | | |
|---|------|---|-----|
| (d) Find V by mesh analysis method such that the current through the 10 volt source is zero. Fig. (3) | (04) | 1 | BL2 |
|---|------|---|-----|



- | | | | |
|--|------|---|-----|
| Q.2 (a) Draw the sinusoidal waveforms of voltage and current for the following condition – | (02) | 2 | BL1 |
| (i) Current lags the voltage (ii) Current leads the voltage. | | | |
| (b) Give the reason, why power factor should be maintained near to unity? | (02) | 2 | BL2 |

- (c) In a series RLC circuit, the maximum inductor voltage is twice the capacitor voltage maximum. However, the circuit current lags the applied voltage by 30° and the instantaneous drop across the inductance is given by $v_L = 200 \sin 377t$ volt. Assuming the resistance being 50Ω , find the values of the inductance and capacitance. (04) 2 BL3
- OR**
- (d) The applied voltage in a parallel RLC circuit is given by (04) 2 BL3

$$v = 100 \sin (5000t + (\pi/4)) \text{ volt}$$
 If the values of R, L and C be given as 25Ω , 2mH and $25\mu\text{F}$, find the total current supplied by the source.
- 2.3 (a) Draw the hysteresis curve for ferromagnetic material with proper labeling. Also mark coarcivity and retentivity on the curve. (02) 3 BL1
 (b) Derive EMF equation of a transformer. (02) 3 BL2
 (c) A magnetic circuit comprises three parts in series, each of uniform cross-sectional area. They are: (04) 3 BL3
 (i) a length of 100mm and cross-sectional area 50mm^2 ,
 (ii) a length of 80mm and cross-sectional area 90mm^2 ,
 (iii) an airgap of length 0.5mm and cross-sectional area 150mm^2 .
 A coil of 5000 turns is wound on part (ii), and the flux density in the airgap is 0.5T . Assuming that all the flux passes through the given circuit, and that the relative permeability is 1500 , estimate the coil current to produce such a flux density.
- OR**
- (d) The primary and secondary windings of a 100kVA transformer have resistance of 0.5Ω and 0.002Ω , respectively. The primary and secondary voltages are 11kV and 400V , respectively and the core loss is 3kW , assuming the power factor of the load to be 0.85 . Calculate the efficiency on (i) full load and (ii) half load. (04) 3 BL3
- 2.4 (a) Enlist 3- ϕ power measurement methods. Draw the circuit of any one method. (02) 4 BL1
 (b) Define the terms – prime mover and synchronous generator. Also write their applications. (02) 4 BL1
 (c) A balanced delta-connected 3- ϕ load is fed from a 3- ϕ , 400V supply. The line current is 25A and the total power absorbed by the load is 10kW . Calculate – the impedance in each branch, the power factor and total power consumed if the same impedances are star connected. Also draw circuit for given condition.. (04) 4 BL3
- OR**
- (d) A balanced, 3- ϕ , star-connected load is fed from a 400V , 3- ϕ , 50Hz supply. The current per phase is 25A (lagging), and the total active power absorbed by the load is 15kW . Determine – the resistance and inductance of the load per phase, the total reactive power, the total apparent power. (04) 4 BL3
- 2.5 (a) Enlist losses occur in the rotating machines and define them. (02) 5 BL1
 (b) Draw the schematic diagram of capacitor start capacitor run 1- ϕ induction motor. Also write application of it. (02) 5 BL1
 (c) Explain armature voltage control method of speed control of DC motor using suitable diagram. (Answer in brief) (04) 5 BL2
- OR**
- (d) Explain operation of 3- ϕ induction motor using suitable diagram. (Answer in brief) (04) 5 BL2

~~DECEN~~ ^{Feb effective} 2023 EXAMINATION

I B. E./B.Tech. (4 YDC)

ME 10049/10149: Engineering Graphics

Time 3 Hours]

[Max. Marks:70]

**All questions are compulsory . Use first angle projection method unless otherwise stated.
Assume suitable dimensions if necessary.**

- | | | | | Marks | CO | BL | PI | | |
|------------|-----|---|------|-------|----|--------|----|--|--|
| Q.1 | (a) | The distance between Coimbatore and Madurai is 200 km and its equivalent distance on map measures 10 cm. Draw the diagonal scale and indicate 223 km and 135 km on the scale. | (7) | 1 | 2 | 10.1.2 | | | |
| | (b) | Draw an involute of a pentagon having each side of 30 mm length. Draw a normal and tangent to this curve at a point P situated at a distance 160 mm from the endpoint of the curve. | (7) | 1 | 2 | 1.2.1 | | | |
| OR | | | | | | | | | |
| | (c) | A circle of 50 mm diameter rolls on inner side of the circumference of another circle of 175 mm. Trace the locus of a point on the circumference of the rolling circle for one complete revolution. Name the curve traced. | (7) | 1 | 2 | 1.2.1 | | | |
| Q.2 | (a) | The top view of a 75 mm long line CD measures 50 mm. Its one end A is 50 mm in front of V.P. and 15 mm below H.P. D is 15 mm in front of V.P. Draw the front view of CD and determine its inclinations with the H.P. and the V.P. Show its traces also. | (7) | 2 | 3 | 2.2.2 | | | |
| and | | | | | | | | | |
| | (b) | The top view of a square lamina of sides 60 mm is a rectangle of sides 60 mm x 20 mm. with the longer side of the rectangle being parallel to the XY line in both, the front view and the top view of the lamina. Draw its projections. | (7) | 2 | 3 | 2.1.3 | | | |
| OR | | | | | | | | | |
| | (c) | A thin circular plate of 50 mm diameter is resting on a point A of its rim with the surface of the plate inclined at 45° to the H.P. and the diameter line through A inclined at 30° to the V.P. Draw the projections of the plate. | (14) | 2 | 3 | 2.3.1 | | | |
| Q.3 | | A cylinder with 50 mm diameter of its base and axis 70 mm long has its axis inclined at 30° to the V.P. and the front view of the axis is inclined at 30° to the line XY. Draw the projections of the cylinder. | (14) | 3 | 2 | 2.3.1 | | | |
| OR | | | | | | | | | |
| | | A hexagonal pyramid with 25 mm edges of base and axis 50 mm long, is resting in H.P. on one of its triangular faces with its axis parallel to the V.P. It is cut by a section plane perpendicular to the H.P., inclined at 30° to the V.P. and passing through a point P on | (14) | 3 | 2 | 2.3.1 | | | |

the axis 20 mm from the base. Draw the top view, sectional front view and true shape of the section.

- Q.4 (a) Draw an isometric scale for a distance of 70 mm. having 7 equal divisions. When is this type of scale used ? (4) 4 1 10.1.1
- (b) The projections of a solid composed of a truncated half-cylinder and a cut half-prism are given in figure -1. Draw the development of its lateral surface. Radius of cylindrical portion 25 mm. (10) 4 3 2.2.3

OR

- (c) Draw the isometric view of the solid whose orthographic projections are given in figure-2 (10) 4 2 10.1.1

- Q.5 Draw the (a) Front view (b) Top view and (c) Left hand side view (6+4+ 5 3 10.1.2 4)

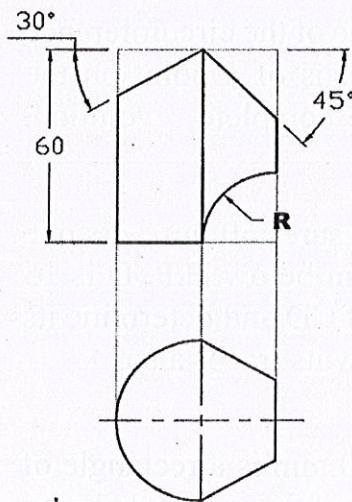


Figure-1

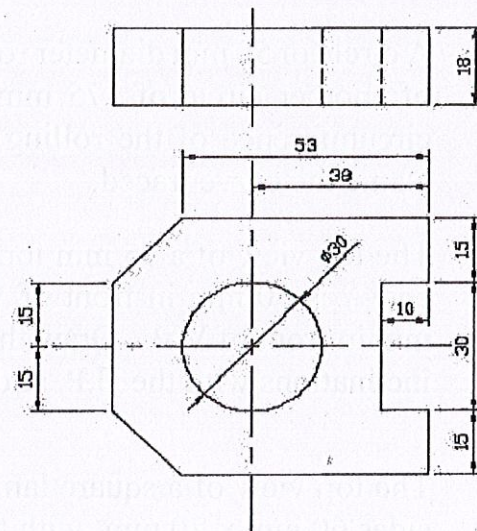


Figure-2

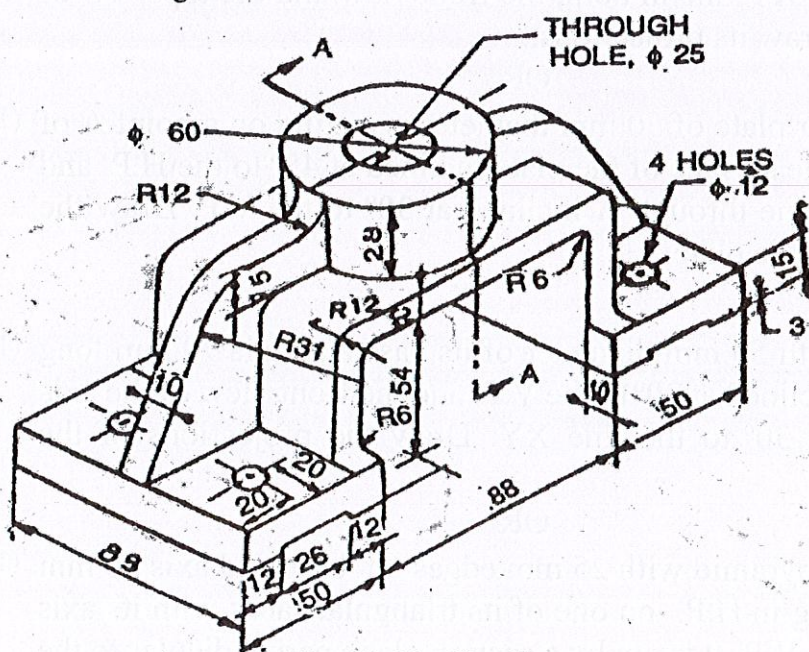


Figure-3

APRIL 2022 EXAMINATION
I B. Tech. (4 YDC)
ME10149: Engineering Graphics

856

[Time 3 Hours]

[Max. Marks:70]

All questions are compulsory.

Use first angle projection method unless otherwise stated.

Assume suitable dimensions if necessary.

		Marks	CO	BL	PI
Q.1	Solve any two:				
(a)	The distance between Parry's corner and Egmore is 2.5 km. On inspection of the road map, its equivalent distance measures 5 cm. draw a diagonal scale to read minimum 50m. Show on it a distance of 6350m.	(7)	1	2	1.4.1
(b)	Draw a rectangular hyperbola when the position of a point P on the curve is at a distance of 35 mm and 50 mm from two asymptotes. Draw a tangent to it at a point 40 mm away from any one asymptote	(7)	1	3	1.3.1
(c)	A circle of 50 mm diameter rolls along a straight line without slipping. Draw the curve traced out by a point P on the circumference, for one complete revolution of the circle. Name the curve. Draw a tangent to the curve at a point on it 40 mm from the line.	(7)	1	3	1.3.1
Q.2					
(a)	The projectors of the ends A and B of a line AB are 120 mm apart. A is 55 mm. below H.P. and 55 mm. behind V.P., B is 100 mm. above H.P. and 20 mm. in front of V.P. Determine the true length , traces and inclinations of the line AB. Measure the distance of each trace from XY.	(7)	2	3	3.2.1
	and				
(b)	A thin 30° - 60° set-square has its longest edge in the V.P. and inclined at 30° to the H.P. Its surface makes an angle of 45° with the V.P. Draw its projections.	(7)	2	3	10.1.3
	OR				
(c)	A cylinder of 50 mm base diameter and 70 mm long axis has a point of its base circle in the V.P. Its axis is inclined at 30° to V.P. and 45° to the H.P. Draw its projections.	(14)	2	3	2.1.3
Q.3					
(a)	Draw the projections of a cone resting on the ground on its base and show on them, the shortest path by which a point P, starting from a point on the circumference of the base and moving around the cone will return to the same point. Base of cone is 60 mm diameter; axis 80 mm long.	(14)	3	3	2.3.1
	OR				
(b)	A right regular pentagonal pyramid, 40 mm side of base and 100 mm height of axis, lies on one of its slant faces in H.P. with its axis parallel	(14)	3	3	10.1.3

to V.P. An auxiliary plane perpendicular to H.P. and inclined at 30° to V.P. cuts the pyramid and passes through a point 47 mm from the base along the axis. Draw sectional elevation and the true shape of the section.

- Q.4 Draw the Front View, Top View and Both Side Views of the object (14) shown in Figure-1.

4 3 2.4.4

- Q.5 (a) Draw the isometric projection of the block shown in Figure-2 (14)

4 3 2.2.4

OR

- (b) Draw the isometric projection of a sphere of diameter 50 mm., when it is placed centrally on the frustum of a square pyramid of sides 50 mm of the top and 70 mm sides of base and height 30 mm.

4 3 3.2.3

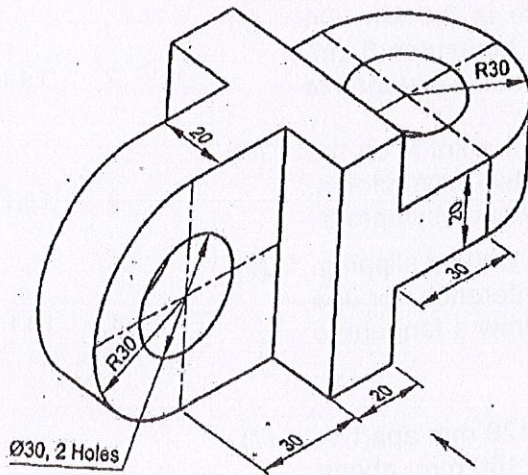


Figure-1

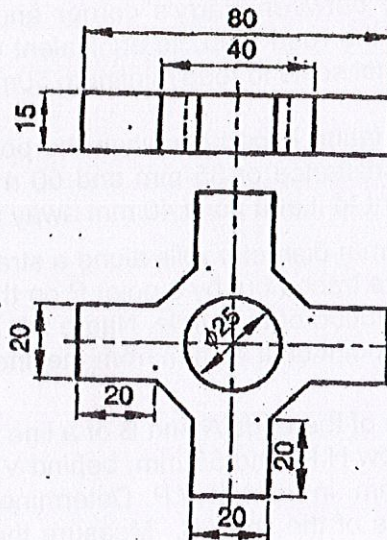


Figure-2

Exam Code: 2126

February 2022 Examination
I-Yr., B.Tech.
ME10149: Engineering Graphics

[Max. Marks : 40]

PART-B

QB. Solve the following 5 questions each carry equal of 8 marks each.

S.No	Questions	Marks	CO	BL	PI
Q1	(a) Draw Plan, Elevation and Left hand side view of fig. 1 shown herewith OR (b) Draw isometric projection of a cylinder of diameter 40 mm and axial length of 70 mm lying on the HP, when its axis is parallel to both HP and VP.	8	CO1	L2	1.3.1
Q2	(a) In a map a 36 km distance is shown by a line 45 cm long. Calculate the RF and construct a plain scale to read kilometers and hectometers, for max. 12 km. Show a distance of 8.3 km on it. OR (b) Draw an ellipse by concentric circle method, take major axis as 100 mm and minor axis as 70 mm long.	8	CO2	L3	1.1.18
Q3	(a) The end A of a line AB is 15 mm above the HP and 10 mm in front of VP. The other end B is 45 mm above the HP and 5 mm in front of VP. The distance between the end projectors is 38 mm. Draw the projection of the Line AB and find its true length. OR (b) Draw the projection of a pentagonal Prism, base 25 mm side and axis 50 mm long resting on one of its rectangular face on the HP, with the axis inclined at 45° to the VP.	8	CO3	L4	1.1.2
Q4	(a) A right square pyramid of base side 40 mm and height 120 mm is resting on the ground on its base, it is bisected by a horizontal cutting plane. Draw the development of the lower part of this Pyramid. OR (b) A right circular cone of base diameter 50 mm and height 80 mm is bisected by an inclined plane making an angle of 45° to the HP. Draw its projection, and show sectional top view	8	CO4	L2	5.1.2
Q5	(a) Write AutoCAD command for drawing the geometric entities: (i) Circle, (ii) Arc, (iii) Ellipse and (iv) Polylines OR (b) For drawing the projection of a square Prism of side 40 mm and height 60 mm resting on its base on the HP. Write various AutoCAD commands for this projection	8	CO5	L4	1.1.2

FEB 2023 EXAMINATION
B.E. /B.TECH EXAM
MA 10001:-MATHEMATICS-I

Time: 3 Hrs.]

[Max. Marks: 70

TOTAL NO. OF QUESTIONS IN THIS PAPER:5

Note: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

S. No.	Questions	Marks	CO	BL	PI
Q.1	(a) Write down the statement of Taylor's theorem for one variables.	(02)	CO1	1	1.1.1
	(b) If $u = x(1-y)$, $v = xy$ then find the value of $\frac{\partial(u,v)}{\partial(x,y)}$.	(02)	CO1	2	1.1.1
	(c) If $z = f(x, y)$, $x = e^u + e^{-v}$, $y = e^{-u} - e^v$ then show that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$	(03)	CO1	2	1.1.1
	(d) Expand $e^{ax} \sin by$ in powers of x & y as far as the term of third degree using Maclaurin's theorem.	(07)	CO1	3	1.1.1
	OR				
	(e) If $u = \sin^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right]$ then prove that (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{-\sin u \cos 2u}{4 \cos^3 u}$	(07)	CO1	3	1.1.1
Q.2	(a) Define curvature of a curve.	(02)	CO2	1	1.1.1
	(b) Write down the steps to find maximum value for any function of two variables.	(02)	CO2	1	1.1.1
	(c) Find the radius of curvature for the curve $y = \frac{a}{2}(e^{x/a} + e^{-x/a})$ at the point (x, y) .	(03)	CO2	2	1.1.1
	(d) A Rectangular box, open at the top, is to have a volume of 32 cc. Find the dimensions of the box requiring least material for its construction.	(07)	CO2	4	1.1.1
	OR				
	(e) Find the asymptotes of the curve $x^3 + 2x^2y - xy^2 - 2y^3 + xy - y^2 - 1 = 0$	(07)	CO2	2	1.1.1
Q.3	(a) Prove that $\int_0^1 \frac{x^2}{\sqrt{1-x^5}} dx = \frac{1}{5} \beta\left(\frac{3}{5}, \frac{1}{2}\right)$	(02)	CO3	2	1.1.1
	(b) Evaluate $\int_0^2 \int_0^2 (x^2 + y^2) dx dy$	(02)	CO3	2	1.1.1

	(c)	Evaluate $\int_0^{\infty} \sqrt{x} e^{-3\sqrt{x}} dx$	(03)	CO3	2	1.1.1
	(d)	Evaluate the following integral by changing the order of integration $\int_{x=0}^{2a} \int_{y=\frac{x^2}{4a}}^{3a-x} dx dy$	(07)	CO3	3	1.1.1
		OR				
	(e)	Evaluate $\iint_R \frac{r dr d\theta}{\sqrt{a^2 + r^2}}$ over one loop of the curve $r^2 = a^2 \cos 2\theta$	(07)	CO3	3	1.1.1
Q.4	(a)	Find the maximum value of r for the curve $r = a + b \cos \theta, a < b$	(02)	CO4	2	1.1.1
	(b)	Find point of intersection with coordinate-axis for the curve $y = x(x^2 - 1)$.	(02)	CO4	2	1.1.1
	(c)	Find the area bounded by the parabola $y = x^2$ and the line $y = x$	(03)	CO4	3	1.1.1
	(d)	Find the volume of the solid formed by the revolution of the cardioid $r = a(1 + \cos \theta)$ about initial line.	(07)	CO4	4	1.1.1
		OR				
	(e)	Find the whole length of the curve $8a^2 y^2 = x^2(a^2 - x^2)$	(07)	CO4	3	1.1.1
Q.5	(a)	In Boolean Algebra $[B, +, \cdot, ']$ prove that $a + 1 = 1$.	(02)	CO5	2	1.1.1
	(b)	In Boolean Algebra $[B, +, \cdot, ']$ prove that $a.b + a'.b' = (a' + b)(a + b')$.	(02)	CO5	2	1.1.1
	(c)	In Boolean Algebra $[B, +, \cdot, ']$ simplify the following $[a + (a'.b)'][a' + (a.b)]$	(03)	CO5	2	1.1.1
	(d)	(i) Replace the given function by a simple switching circuit. $F(x, y, z) = (x + y)(x + z) + z.(x + y.z)$ (ii) State and prove Involution law in Boolean algebra.	(04+03)	CO5	3	1.1.1
		OR				
	(e)	Change the Boolean function $F(x, y, z) = (x + y + z).(xy + x'z)$ into conjunctive and disjunctive normal form.	(07)	CO5	2	1.1.1

JUNE-JULY 2022 EXAMINATION

B.E. /B.TECH EXAM

MA 10001:-MATHEMATICS-I

Time: 3 Hrs.]

[Max. Marks: 70

TOTAL NO. OF QUESTIONS IN THIS PAPER:5

e: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

S. No.	Questions	Marks	CO	BL	PI
Q.1	(a) Write down the statement of Maclurin's theorem for function of two variables.	(02)	CO1	1	1.1.1
	(b) If $u = e^x \sin y, v = x + \log \sin y$ then find the value of $\frac{\partial(u,v)}{\partial(x,y)}$.	(02)	CO1	1,2	1.1.1
	(c) If $u = (1 - 2xy + y^2)^{-1/2}$ then show that $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = u^3 y^2$.	(03)	CO1	2	1.1.1
	(d) Find the first four terms in the expansion of $\log_e \sin(x+h)$ in the ascending powers of h by Taylor's theorem. Hence find the value of $\log_e \sin 31^\circ$ to four places of decimal.	(07)	CO1	3	1.1.1
	OR				
	(e) If $u = \sin^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right]$ then prove that i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$ ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{\sin u \cos 2u}{4 \cos^3 u}$	(07)	CO1	2	1.1.1
Q.2	(a) Find the radius of curvature of the curve $s = c \tan \psi$. at the point (s, ψ) .	(02)	CO2	2	1.1.1
	(b) Find the stationary point of the function $f(x, y) = 4x^2 + 9y^2 - 8x - 12y + 4$	(02)	CO2	1,2	1.1.1
	(c) Find the co-ordinate of center of curvature of the parabola $y^2 = 4ax$ at the point $(a, 2a)$.	(03)	CO2	2	1.1.1
	(d) A rectangular box open at the top is to have a volume of 32 cc. find the dimension of the box requiring least material for its construction.	(07)	CO2	3	1.1.1
	OR				
	(e) Find the asymptotes of the curve $4x^3 - x^2y - 4xy^2 + y^3 + 3x^2 + 2xy - y^2 - 7 = 0$	(07)	CO2	2	1.1.1
Q.3	(a) Evaluate $\int_0^\infty x^n e^{-\sqrt{ax}} dx$	(02)	CO3	2	1.1.1

	(b)	Prove that $\int_0^1 \frac{x^2}{\sqrt{1-x^5}} dx = \frac{1}{5} \beta\left(\frac{3}{5}, \frac{1}{2}\right)$	(02)	CO3	2	1.1.1
	(c)	Evaluate $\int_0^4 \int_0^x x^2 dy dx$.	(03)	CO3	2	1.1.1
	(d)	Evaluate the following integral by changing the order of integration $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x dy dx}{\sqrt{x^2+y^2}}$	(07)	CO3	2	1.1.1
		OR				
	(e)	Evaluate $\iint_R r^3 dr d\theta$ over the area included between the circles $r = 2a \cos \theta$ & $r = 2b \cos \theta, b < a$.	(07)	CO3	2	1.1.1
Q.4	(a)	write down the formula for surface area of the solid generated by revolution about the x-axis, of the area bounded by the curve, the x-axis and the ordinates $x = a, x = b$.	(02)	CO4	1	1.1.1
	(b)	Find the symmetry of the curve $y^2(a^2 + x^2) = x^2(a^2 - x^2)$ and also find point of intersection with x-axis.	(02)	CO4	1	1.1.1
	(c)	Find the area included between the parabola $y^2 = 4ax$ and its latus rectum.	(03)	CO4	2	1.1.1
	(d)	Find the volume of the solid formed by the revolution of the curve $xy^2 = 4(2-x)$ through four right angles about the y-axis.	(07)	CO4	3	1.1.1
		OR				
	(e)	Find the whole perimeter of the cardioids $r = a(1 + \cos \theta)$	(07)	CO4	2	1.1.1
Q.5	(a)	Define minimal Boolean function with example.	(02)	CO5	1	1.1.1
	(b)	In Boolean Algebra $[B, +, \cdot, ']$ prove that $a + a = a$.	(02)	CO5	1	1.1.1
	(c)	In Boolean Algebra $[B, +, \cdot, ']$ prove that $(x + y \cdot z)(y' + x)(y' + z') = x \cdot (y' + z')$	(03)	CO5	2	1.1.1
	(d)	Replace the switching function $F(a, b, c) = a \cdot c + b \cdot (b' + c) \cdot (a' + b \cdot c')$ by a simple switching circuit and draw the circuit for the same.	(07)	CO5	3	1.1.1
		OR				
	(e)	Change the Boolean function $F(x, y, z) = (x + y) \cdot (x + z') + (y + z')$ into disjunctive normal form.	(07)	CO5	2	1.1.1

Time : 90 Minutes]

[Max. Marks :40

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note: Each question carry three subparts a, b and c. Attempt any two from a , b and c. All questions carry equal marks.

Questions		Marks	COS	BL	PI
Q.1 (a)	(i) If $u = \tan^{-1}\left(\frac{y^2}{x}\right)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u$	04	CO1	1,2	1.1.
	(ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\sin^2 u \sin 2u$				
(b)	If $u = x + y^2$, $v = y + z^2$, $w = z + x^2$ prove that $\frac{\partial x}{\partial u} = \frac{1}{1+8xyz}$	04	CO1	1,2	1.1.
(c)	Expand $e^x \cos y$ in power of x & y as far as the terms of the third degree.	04	CO1	1,2	
Q.2 (a)	A rectangular box is placed on xy plane whose vertex is at the origin. Find the maximum volume of the box if the vertex facing to the origin lies on the plane $6x + 4y + 3z = 24$	04	CO2	1,2	1.1.
(b)	Find the equation of the circle of curvature of the rectangular hyperbola $xy = 12$ at the point $(3, 4)$	04	CO2	1,2	1.1.
(c)	Find the asymptote of the curve $(2x - 3y + 1)^2(x + y) - 8x + 2y - 9 = 0$	04	CO2	1,2	1.1.
Q.3 (a)	Show that $\int_0^2 x(8 - x^3)^{\frac{1}{3}} dx = \frac{16\pi}{9\sqrt{3}}$	04	CO3	2,3	1.1.
(b)	Transform the integral $\int_0^a \int_0^{\sqrt{a^2-x^2}} y^2 \sqrt{x^2 + y^2} dx dy$ by changing to polar coordinate and hence evaluate it.	04	CO3	2,3	1.1.
(c)	Change the order of integration in $\int_0^1 \int_{e^x}^e \frac{dx dy}{\log y}$ and hence evaluate the same	04	CO3	1,2	1.1.
Q.4 (a)	The cardioid $r = a(1 + \cos \theta)$ revolves about the initial line. Find the volume of the solid generated.	04	CO4	1,2	1.1.

(b)	Find the length of the curve of the semicubical parabola $ay^2 = x^3$ to the point (a, a)	04	CO4	1,2	1.1.
(c)	Trace the curve $y^2(x^2 + y^2) + a^2(x^2 - y^2) = 0$	04	CO4	1,2	1.1.
Q.5 (a)	Change the following Boolean function into disjunctive normal form $f(x, y, z) = [x + (x' + y)'] \cdot [x + (y' \cdot z')']$ is	04	CO5	1,2	1.1.
(b)	Draw a circuit for the following Boolean function and replace it by a simpler one $f(x, y, z) = x \cdot z + [y \cdot (y' + z) \cdot (x' + x \cdot z')]$	04	CO5	1,2	1.1.
(c)	Prove the following: (i) $[a + (a' + b)'] \cdot [a + (b' \cdot c)'] = a$ (ii) $(a + b') \cdot (a' + b) \cdot (a' + b') = a' \cdot b'$	04	CO5	1,2	1.1.

FEBRUARY 2023 EXAMINATION
I YEAR B.E./B.TECH EXAMINATION
PH-10016: PHYSICS

1213

TIME: 3 Hrs.

Max. Marks: 70

Note: Total questions are 5. Parts (a), (b) and (c) are compulsory. Attempt any one question from (d) or (e) section.

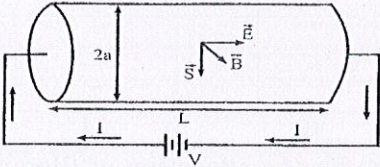
S.No	Questions	Marks	CO	BL	PI
Q.1.					
(a)	A progressive sinusoidal wave is represented by $Y(x, t) = A \sin [(0.2 \text{ m}^{-1})x - (0.4 \text{ s}^{-1})t + \pi/6]$, where x and t are in meter and second respectively. Determine the speed of propagation of the wave.	(2)	1,4	2,3	1.6
(b)	Explain the physical significance of Gauss and Stoke's theorems.	(2)	2,4	1,2	1.5
(c)	A particle with a proper lifetime of $1 \mu\text{s}$ moves through the laboratory at $2.7 \times 10^8 \text{ ms}^{-1}$. (i) What is its lifetime, as measured by observers in the laboratory? (ii) What will be the distance traversed by it before disintegrating?	(3)	1,4	3,5	1.2
(d)	What are Maxwell's equations? State and Derive Maxwell's equations in differential form.	(7)	2,4	1,3	2.3
OR					
(e)	Define Poynting vector. Derive an expression for it and explain its physical significance for electromagnetic wave in free space.	(7)	2,4	2,4	2.2
Q.2					
(a)	In a Newton's rings experiment the diameter of 10 th ring changes from 1.40 to 1.27 cm when a drop of liquid is introduced between the lens and the glass plate. Calculate the refractive index of the liquid.	(2)	2,4	2,5	1.6
(b)	A monochromatic light of wavelength 650 nm falls normally on a grating. The first order spectrum is observed at 15° from the mean position. Calculate the grating element.	(2)	3,4	3,5	1.2
(c)	Define resolving power of an optical instrument? Explain Rayleigh's criterion for limit of resolution.	(3)	1,3	1,2	1.4
(d)	Describe and explain the formation of Newton's rings in reflected light. Prove that in reflected light the diameter of the dark rings is proportional to the square roots of natural numbers.	(7)	2,3	2,3	1.5
OR					
(e)	Discuss the phenomenon of Fraunhofer's diffraction at a single slit and determine the conditions for maximum and minimum.	(7)	2,3	3,4	1.2
Q.3					
(a)	The lifetime of an excited state of an atom is about 10^{-8} sec. calculate the minimum uncertainty in the determination of the energy of the excited state.	(2)	1,4	2,4	1.6
(b)	What are matter waves? Write any four properties of matter waves.	(2)	1,2	2,3	2.2
(c)	Calculate the de Broglie wavelength associated with (i) A 46-gm golf ball with velocity 36 m/s (ii) An electron with a velocity 10^7 m/s Which of these two, show wave character and why?	(3)	2,4	2,4	1.2

Time: 180 min

Total Number of Questions: 5

Max. Marks: 70

NOTE: Attempt any TWO part of each question. Each part carry 7 marks.

Questions	CO	BL	PI
1a.- State Poynting theorem. With the help the given figure drive the expression for energy per unit time ($\vec{S}$) passing through the surface of the wire. Symbols have their usual meanings. (03+04) 	1,4	3	2.1.2
1b.- Prove the following (i) Electromagnetic wave have transverse nature. (02) (ii) $x^2 + y^2 + z^2 = c^2 t^2$ is invariant under Lorentz transformation. (03) (iii) Velocity of light in free space = $1/\sqrt{\mu_0 \epsilon_0}$ (Hint: Use Maxwell's equation) (02)	2,4	2	1.3.1
1c.- State fundamental postulates of special theory of relativity and deduce Lorentz transformation of space and time. (03+04)	3,4	3	1.3.1
2a.- Explain the theory of plane transmission grating for gating n^{th} order maxima. Also, compare the diffraction pattern of grating with that of a single slit. (05+02)	1,2	4	2.2.4
2b.- Define resolving power, limit of resolution, Rayleigh criteria and calculate the (03) (i) angular position of 4^{th} order maxima, if for the normal illumination of a transmission grating, the first order diffraction maxima is at 12°. (02) (ii) diameter of a telescope lens, if at 600nm the required resolution is $0.1''$. (02)	3,4	5	2.3.1
2c.- For sustained interference we need two coherent sources. With the help of necessary theory and diagram demonstrate the application of interference in laboratory. However you have a sodium vapor lamp. (07)	1,4	3	4.1.3
3a.- Discuss Heisenberg uncertainty principle and explain the following. (03) (i) Quantization of energy level of an atom. (02) (ii) Non existence of electron in nucleus. (02)	1,2	2	1.2.1
3b.- What is Compton effect ? Why the effect is significant only for microscopic particles ? Develop the expression for Compton shift and angle of recoil. (01+01+05)	1,2	3	2.3.2
3c.- Deduce eigenfunction and eigenvalues of an electron in a deep potential well of width L. Find the probability of finding the electron in first excited state between $0.45L$ and $0.55L$ of the well. (04+03)	2,4	5	2.3.1
4a.- Prove that, being the negative temperature state, the population inversion can be achieved at room temperature. Why the two level lasing is not possible ? (04+03)	2,3	2	1.2.1
4b.- Draw energy level transitions in He-Ne laser showing values of each level and lasing wavelengths. Also, for selective wavelength lasing, discuss the significance of He:Ne ratio, tube diameter and length of the optical cavity. (03+04)	2,3	1	2.1.3
4c.- Describe the construction and working of ruby laser. What is the main drawback of ruby laser ? Calculate the number of photon per pulse of ruby laser that emits pulses of 0.15 J for 12 ps and emits light of wavelength 694.4 nm. (04+01+02)	2,4	3	2.1.2
5a.- Briefly describe intermodal and intramodal dispersion in optical fiber. Drive the expression for total time delay due to modal dispersion in step index fiber: (03+04)	1,3	4	3.3.1
5b.- In optical fiber, why cladding is required? If the cut-off angle for light entering a step index fiber ($n_{core} = 1.425$) from air is 8.50° find the values of (02) (i) Numerical aperture (NA), fractional refractive index and n_{clad} of the fiber ? (03) (ii) How the value of these parameter changes when fiber submersed in water ? (02)	1,4	3	2.1.2
5c.- Explain the terms normalized frequency, numerical aperture, acceptance angle and fiber optic communication system. On the basis of mode of fiber and type of index, which fiber will you prefer for long distance communication and why ? (04+03)	2,3	1	5.2.1

Detail of COs, BL and PIs

- Course Outcome (CO) {
1. Knowledge of multiphysics to understand engineering problems
 2. Skills to use logic towards engineering problems with multiphysics implementation.
 3. Ability to use modern engineering physics techniques and tools.
 4. Solve engineering problems with the knowledge of basic and applied physics.

Bloom's Taxonomy Levels (BL): 1-Remembering, 2-Understanding, 3-Applying, 4-Analyzing, 5-Evaluating, 6-Creating

Unit	Remember (1)		Understand (2)		Apply (3)			Analyzing (4)	Evaluate (5)	Create (6)	Tot.
	Recall	Label	Prove	Explain	Use	Develop	Find	Compare	Calculate	Build	
I	-	-	7	-	7	7	-	-	-	-	21
II	-	-	-	-	7	-	-	7	7	-	21
III	-	-	-	7	-	7	-	-	-	-	14
IV	-	7	7	-	-	-	7	-	7	-	28
V	7	-	-	-	-	-	7	7	-	-	21
Tot.	14		21		42			14	14	0	105

Table 1: Table of specifications (TOS) for the allocation of Bloom's taxonomy levels

Program Outcomes (POs)	Performance Indicator (PI)
Engineering knowledge	1.2.1 Apply laws of natural science to an engineering problem 1.3.1 Apply fundamental physics concepts to solve engineering problems
Problem analysis	2.1.2 Identify systems, variables, and parameters to solve the problems 2.1.3 Identify the mathematical, engineering and other relevant knowledge that applies to a given problem 2.2.4 Compare and contrast alternative solution processes to select the best process. 2.3.1 Combine scientific principles and engineering concepts to formulate mathematical models of a system or process that is appropriate in terms of applicability and required accuracy. 2.3.2 Identify assumptions (mathematical and physical) necessary to allow modeling of a system at the level of accuracy required.
Design / Development of Solutions	3.3.1 Apply formal decision making tools to select optimal solutions for further development
Conduct investigations of complex problems	4.1.3 Apply appropriate instrumentation and/or software tools to make measurements of physical quantities
Modern tool usage	5.2.1 Identify the strengths and limitations of tools for (i) acquiring information, (ii) modeling and simulating, (iii) monitoring system performance, and (iv) creating engineering designs.

Table 2: List of program outcomes and the relevant performance indicators

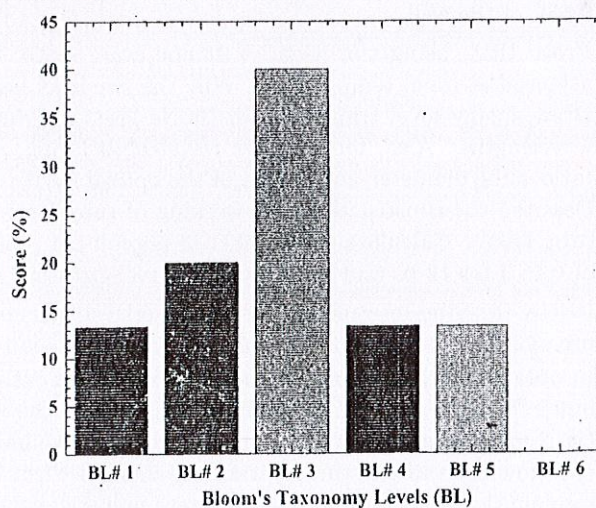
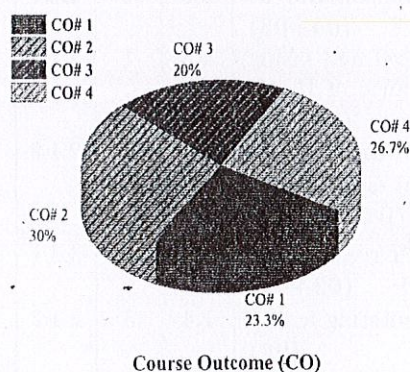


Figure-2: Mapping of COs and BLs with the score covered by the respective questions.

ONLINE FEB 2022 EXAMINATION I YEAR B.E./B.TECH EXAM
PH-10006/PH 10002PHYSICS

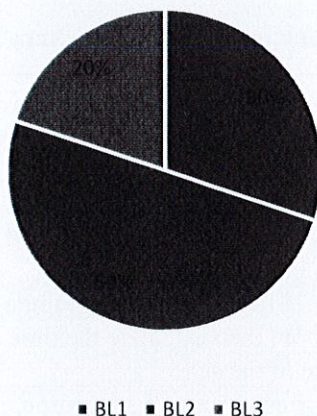
TIME: 90 mins

Max. Marks: 40

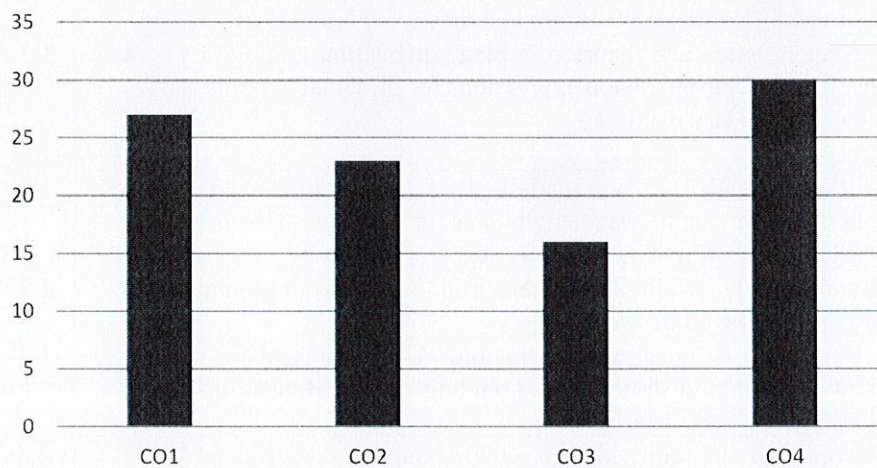
NOTE: All questions are compulsory, each question carry 8 marks.

S.No	Questions	Marks	CO	BL	PI
Q1(a)	Draw a sketch of a plane electromagnetic wave propagating along the z-direction. Depict clearly the direction of electric and magnetic fields varying sinusoidally with z.	2	1,4	1,	1.2.1
(b)	If radius of the sun be equal to $7 \times 10^8 \text{m}$ and the magnitude of power radiated by it is $3.5 \times 10^{26} \text{W}$, then calculate the time average Poynting's vector at the surface of the sun.	2	2,4	1,2	1.2.1
(c)	Derive the continuity equation from Maxwell's equation.	4	1,4	3	1.2.1
Q2(a)	In a Newton's ring experiment the diameter of the 15 th ring was found to be 0.59cm and that of the 5 th ring was 0.336cm. If the radius of the plano-convex lens is 100cm. calculate the wavelength of the light used.	4	2,4	2	1.2.1
(b)	Deduce the missing orders for a double slit Fraunhofer diffraction pattern, if the slit widths are 0.16mm and they are 0.8mm apart.	2	2,4	2	1.2.1
(c)	Give the construction and theory of a plane diffraction grating of the transmission type and obtain expressions for principal maxima, minima and secondary maxima.	4	1,3	3	1.2.1
Q3(a)	X-rays of wavelength $1.2 \text{\AA}$ are scattered by a carbon target at an angle of 60° . Calculate the wavelength of scattered X-ray. Given: $h=6.6 \times 10^{-34} \text{Js}$, $c=3 \times 10^8 \text{ms}^{-1}$, $m=9.1 \times 10^{-31} \text{Kg}$.	2	2,4	1,2	1.2.1
(b)	Calculate the energy required for an electron to jump from ground state in a potential well of width L	2	1,3	2	1.2.1
(c)	Write Heisenberg's uncertainty principle. Use Heisenberg's uncertainty principle to show that the binding energy of electrons in the hydrogen atom is nearly 15eV	4	1,3	3	1.2.1
Q4 (a)	The wavelength of emission is $6000 \text{\AA}$ and the coefficient spontaneous emission is 10^6 per sec. Determine the coefficient for the stimulated emission.	2	2,4	1,2	1.2.1
(b)	A relative population or Boltzmann ratio of $1/e$ is often is considered representative of the ratio of population in two energy states at room temperature $T=300\text{K}$. Determine the wavelength of radiation emitted at that temperature.	2	1,3	1,2	1.2.1
(c)	Describe the principle and working of He-Ne laser.	4	1,2	3	1.2.1
Q5(a)	Explain the phenomenon of total internal reflection of light. How is it used in fiber optic communications?	2	1,4	1,2	1.2.1
(b)	In an optical fiber, the core material has refractive index 1.6 and refractive index of clad material is 1.3. What is the value of critical angle? Also calculate the value of acceptance cone	2	2,4	2	1.2.1
(c)	Derive an expression for acceptance angle and numerical aperture for step index fiber with the help of suitable diagram.	4	1,3	2	1.2.1

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FEBRUARY 2023 EXAMINATION

I B.E. (4YDC) EXAM

CE 10013 / CE-10003: FUNDAMENTALS OF CIVIL ENGINEERING & APPLIED MECHANICS

Time: 3 Hrs.]

Max. Marks : 70

TOTAL NO. OF QUESTIONS IN THIS PAPER : 6

Note: All questions are compulsory. Use internal choice wherever given. Assume suitable data if missing.

Q.1	(a) Answer any two questions	CO	BL	PI	Marks
	(i) Differentiate between concurrent and collinear forces.	CO1	2	1.1.1	02
	(ii) State Varignon theorem and theorem of resolved parts	CO1	2	1.1.1	02
	(iii) State Perpendicular and parallel axis theorem	CO1	2	1.1.1	02
	(b) Answer any two questions				
	(i) Discuss Hypotenusal Allowance.	CO4	2	1.1.1	02
	(ii) Differentiate between Cumulative and Compensating Errors	CO4	2	1.1.1	02
	(iii) Discuss Reciprocal Ranging.	CO4	2	1.1.1	02
Q.2	(a) A man of mass 75 kg starts descending from the top of a spherical dome of 20 m diameter. If the coefficient of friction between the dome surface and the shoes of the man is 0.3, determine how far he can reach without slipping.	CO1	4	1.2.1	04
OR					
	For the part of a truss obtained by passing a section is shown in Fig.1. Using the equation of moment alone determine the unknown forces f_1 , f_2 and f_3 . Verify the results using remaining equations				
	(b) Analyze the truss shown in Fig. 2. Tabulate the results.	CO1	4	1.2.1	10
Q.3	(a) Derive an expression for centre of gravity of right circular cone.	CO2	3	1.2.1	04
	(b) For the plane lamina shown in Fig. 3 determine the product of inertia.	CO2	3	1.2.1	06
OR					
	For the plane lamina shown in Fig. 4, determine the moment of inertia about the axes passing through the centroid.				
	(c) Derive an expression for moment of inertia for a triangle.	CO2	3	1.2.1	04
Q.4	(a) For the beam shown in Fig. 5 determine the support reactions.	CO3	3	1.2.1	04
	(b) For the beam shown in Fig. 6, draw the SFD & BMD.	CO3	4	1.2.1	06
OR					
	Determine the support reactions for the truss shown in Fig. 7				
	(c) Prove that $T_1/T_2 = e^{\mu\theta}$ with usual notations.	CO2	3	1.2.1	04
Q.5	(a) A 30 m chain was found to be 12 cm too long after chaining a distance of 2500 m. It was found to be 18 cm too long at the end of day's work after chaining a total distance of 6000 m. Find the true distance if the chain was correct before the commencement of the work	CO4	3	1.1.1	02
	(b) To find the width of a building lying on either side of a river, the bearings were taken at each corner (A&B) of building from a point P	CO4	3	1.2.1	04

lying on other side and were found to be 30 and 70. The bearing at another point Q lying on the same side as P was found to be 110. The bearings taken at each corners (A&B) from point Q were found to be 310 and 40. The bearing at P was observed to be 290. Determine the width of building if the horizontal distance between P & Q is 30m.

- (c) The fore bearings of a quadrilateral ABCD are tabulated below. CO4 3 1.2.1 04
Determine the included angles.

Lines	Fore Bearing
AB	50°
BC	120°
CD	230°
DA	290°

OR

A regular pentagon of 10m side has the fore bearing of a line AB as 105°15'. Determine the bearings of all other lines.

- Q.6 (a) List various methods of leveling. Discuss barometric leveling in brief CO5 1 1.1.1 04
(b) The following staff readings were taken successively with a level of CO5 4 1.2.1 06
4.0 m on a continuously sloping ground at a common interval of 25 m. The RL of last point is 552.25. Enter the readings in a level book and determine the reduced level of all points. **Use Rise & Fall method** for computations.
0.605, 1.235, 1.86, 2.575, 0.240, 0.915, 1.935, 2.87, 0.565, 1.825, 2.725.

-----X-----X-----

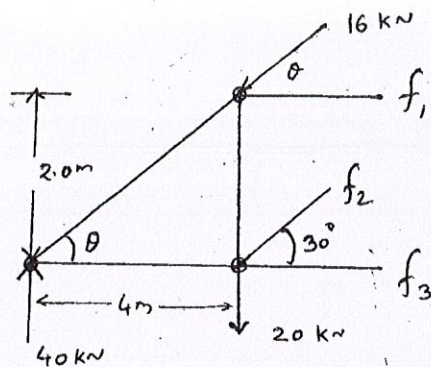


Fig. 1

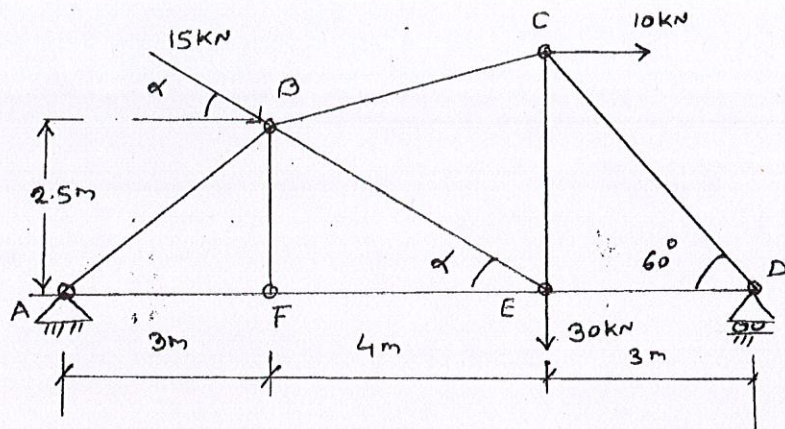


Fig. 2

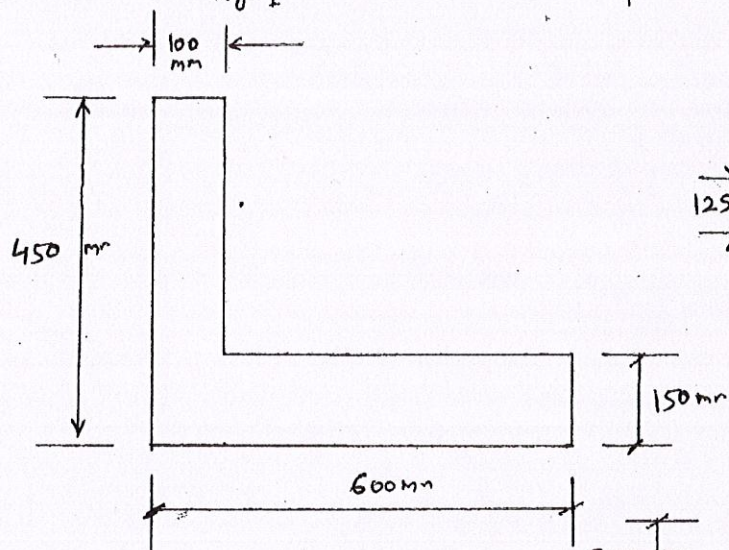


Fig. 3

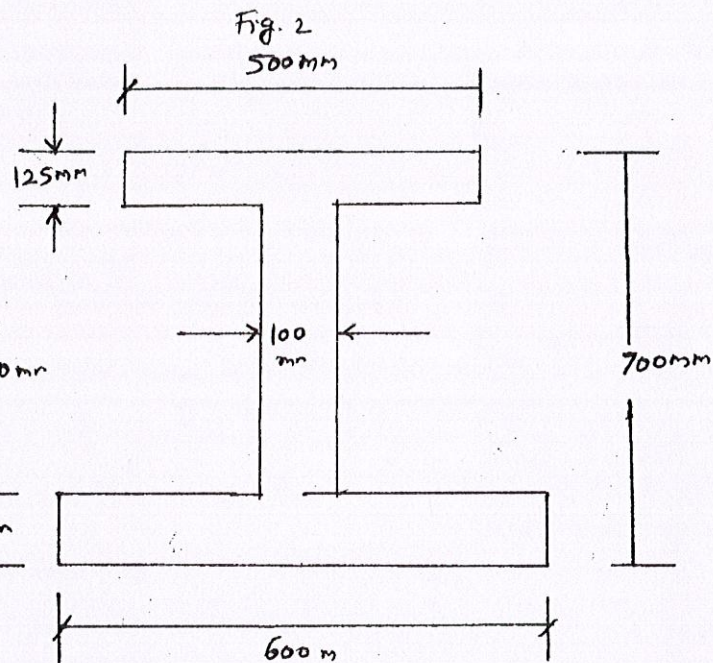


Fig. 4

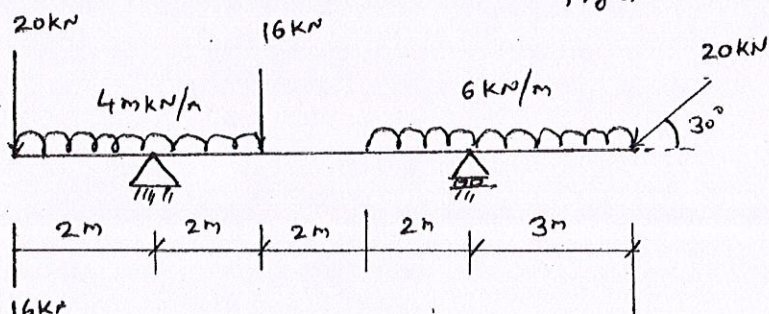


Fig. 5

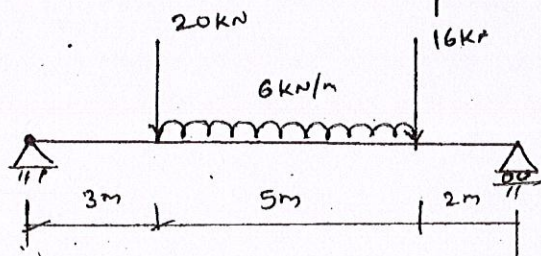


Fig. 6

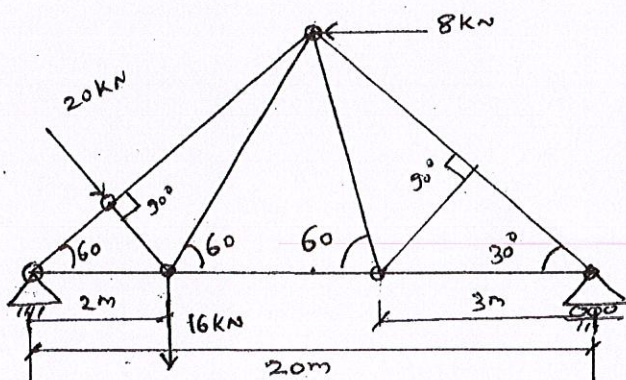


Fig. 7

JUNE – JULY 2022 EXAMINATION
I B. TECH. (4YDC) EXAM
CE-10003: FUNDAMENTALS OF CIVIL ENGINEERING
& APPLIED MECHANICS

Time: 3 Hrs.]

Max. Marks : 70

TOTAL NO. OF QUESTIONS IN THIS PAPER : 6

Note: All questions are compulsory. Use **internal choice** wherever given. Assume **suitable data if missing**.

- Q.1** (a) **Answer any three questions**
- | | | |
|-------------------------------------|---|----|
| (i) | Define force and list various force systems | 02 |
| (ii) | Two forces P & Q are acting over a particle at an angle of 60° , the resultant of the forces is 25 kN making an angle of 45° with the horizontal. Determine forces P & Q. | 02 |
| (iii) | Draw neat diagrams of open & cross belt pulley system of power transmission. | 02 |
| (iv) | State perpendicular & parallel axis theorem. | 02 |
| (b) Answer any two questions | | |
| (i) | Differentiate between error and blunder. | 02 |
| (ii) | List different types of chains | 02 |
| (iii) | Differentiate between prismatic and surveyor's compass (any two) | 02 |
- Q.2** (a) State and prove Lami's theorem. 04
- (b) Analyze the truss shown in Fig.1 and tabulate the results. 08

OR

A ladder 5 m long is supported on wall and floor making an angle of 30° with the floor. The coefficient of friction between the ladder and wall is 0.30 while that between ladder and floor is 0.18. Evaluate how far a person of weight 60 kg can ascend without slippage. If he want to climb the top of ladder what horizontal push is required at the bottom of ladder?

- Q.3** (a) A right circular cone of base width 80 cm and height 240 cm is resting over a hemisphere of 40 cm radius. The density of cone is 1.6 times the density of hemisphere. Determine the centre of gravity. 04
- (b) For the plane lamina shown in Fig.2, determine the moment of inertia about the horizontal axis passing through centroid. 08

OR

For the channel section shown in Fig. 3, determine the product of inertia about the axes passing through the centroid.

- Q.4** (a) Determine the support reactions for the truss shown in Fig. 4. The centre to centre distance between the supports is 30 m. 08
- (b) Two parallel shafts 10 m apart are provided with 800 mm and 300 mm diameter pulleys and are connected by means of a cross belt. The direction of the rotation of the follower pulley is to be reversed by changing over to an open belt drive. How much length of the belt has to be reduced? 04

- Q.5 (a) The initial and final readings of a planimeter used for traversing a circle of 10 cm radius were found to be respectively 5.305 & 8.445. The anchor point was kept outside the figure, and $N = 0$. The figure was traversed again by placing anchor point inside the figure. The initial and final readings were found to be 4.925 & 6.105 respectively, $N = -2$. Now using the same planimeter, the periphery of an unknown figure drawn to a scale of 1:2000 was traced by keeping the anchor point inside the figure. The initial and final readings of the planimeter are 3.255 and 6.250 with $N = -2$. Determine the actual area of the unknown figure in hectares. 04
- (b) The following bearings were observed with a compass in a closed traverse: 08

Line	Fore Bearing	Back Bearing
AB	$16^{\circ}45'$	$198^{\circ}00'$
BC	$224^{\circ}30'$	$47^{\circ}30'$
CD	$207^{\circ}15'$	$25^{\circ}30'$
DE	$67^{\circ}45'$	$247^{\circ}15'$
EA	$155^{\circ}15'$	$332^{\circ}45'$

Determine (i) Included Angles and (ii) Adjust for local attraction

OR

The included angles of pentagon ABCDE are tabulated below. Determine the bearings of all other lines if the bearing of line AB = $242^{\circ}30'$

Included Angle	Magnitude ($^{\circ}$)
A	80
B	70
C	105
D	125
E	160

- Q.6 (a) Discuss profile leveling and cross sectioning. 04
- (b) In order to determine the width of a building lying on the other side of a river, the bearings of the two extreme corners of the building taken from a point P on the side of river are $310^{\circ}30'$ and $40^{\circ}45'$ respectively. Another point Q is taken on the same side of the river and is 50 m away from P. The bearings of the two extreme corners of the building taken from Q are respectively $285^{\circ}15'$ and $54^{\circ}30'$. Determine the width of building. 08

OR

The following readings were taken on a continuously sloping ground by a staff of 3.0 m length. Enter the readings in a page of survey field book. The first reading was taken on a bench mark of RL 300.515 m. Determine the RL of all points. Show all the necessary calculations

0.605, 1.235, 1.86, 2.125, 2.575, 0.235, 0.915, 1.540, 2.135, 2.905, 0.56, 1.525, 2.170 and 2.725.

-----XXXXX-----

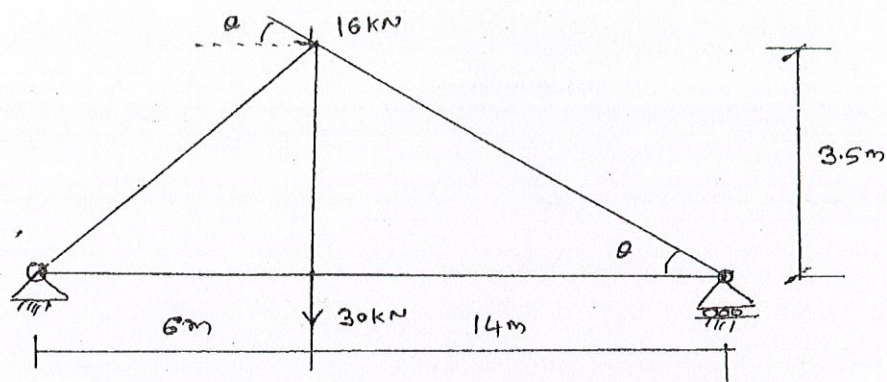


Fig. 1

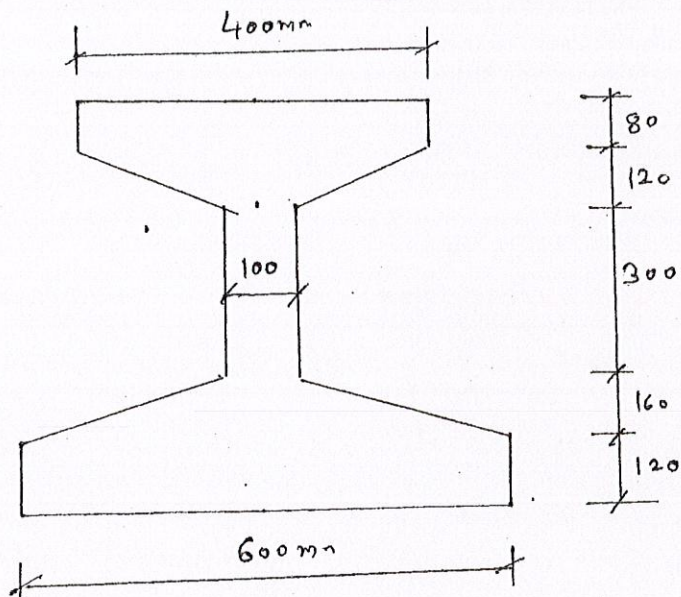


Fig. 2

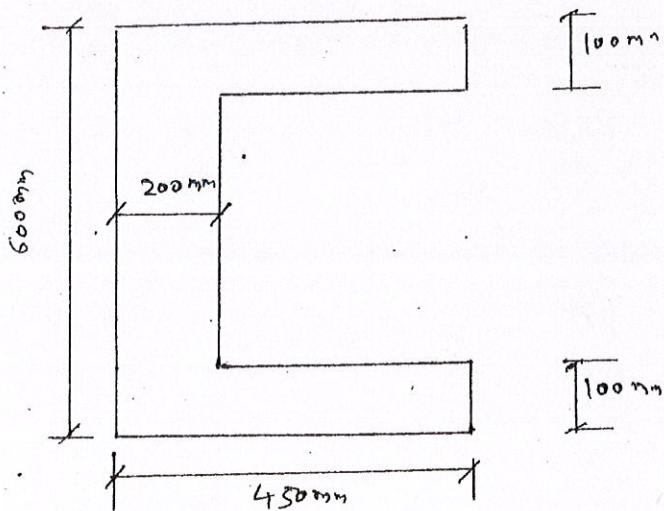


Fig. 3

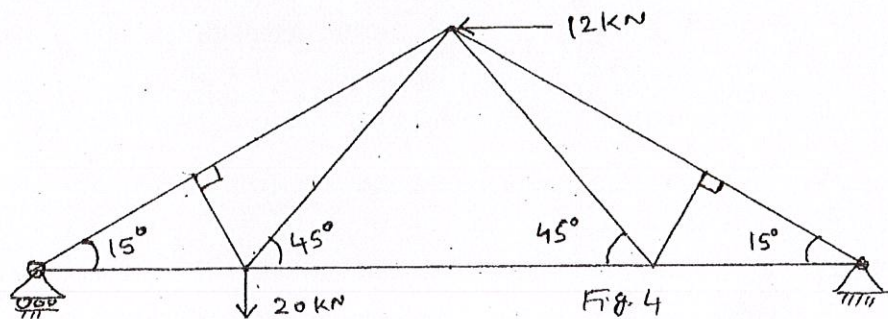


Fig. 4

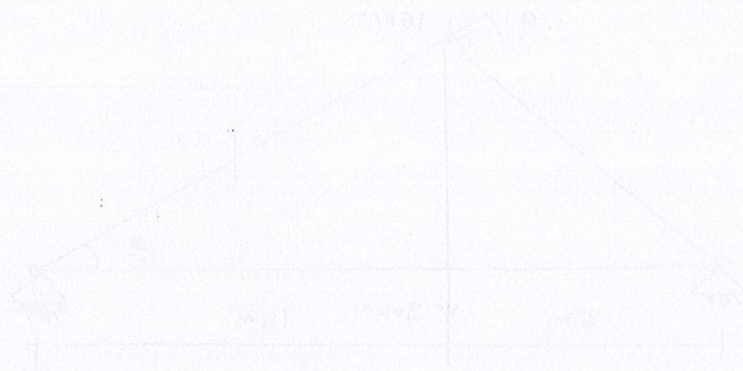


Fig. 1

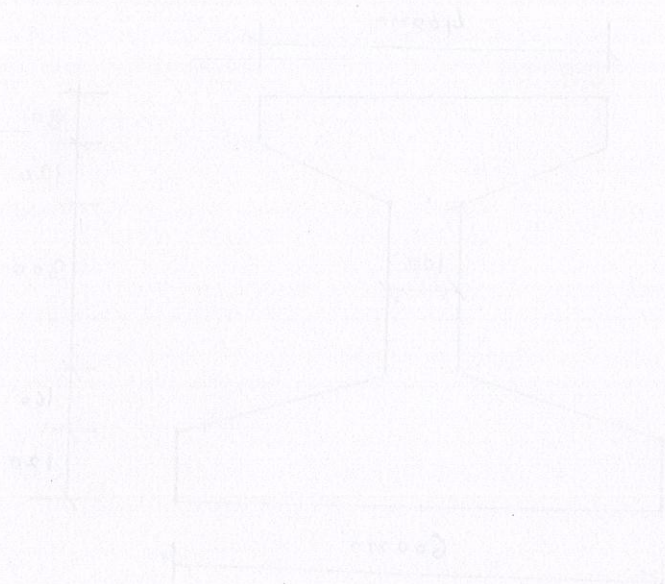


Fig. 2

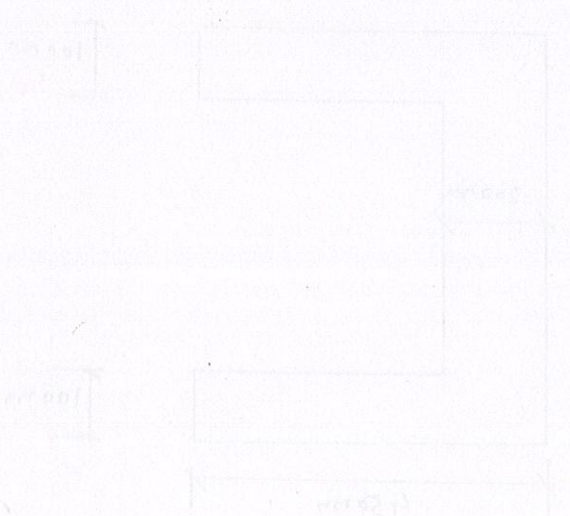


Fig. 3

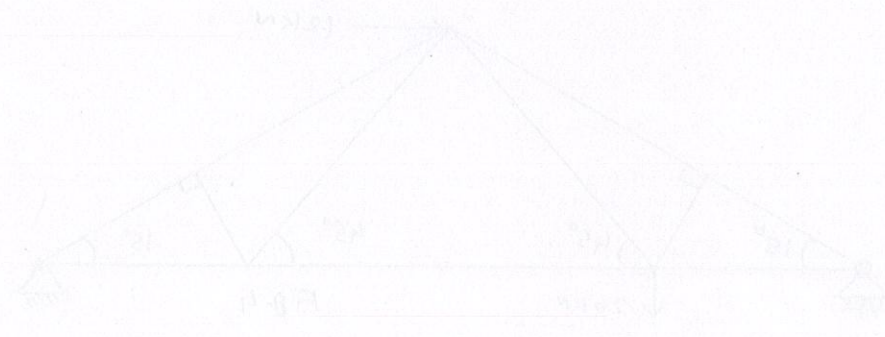


Fig. 4

FEBRUARY 2022 EXAMINATION

I YEAR I SEM B.E/ B.TECH

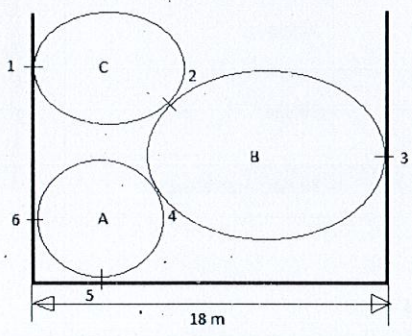
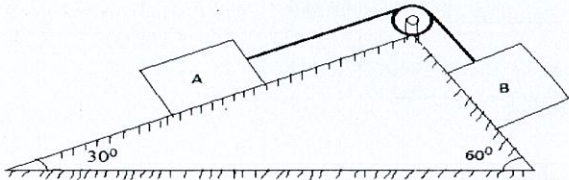
**CE-10003: FUNDAMENTALS OF CIVIL ENGG.AND APPLIED
MECHANICS
PART B**

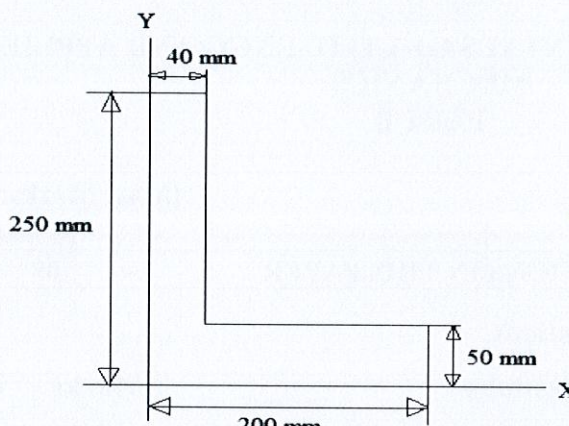
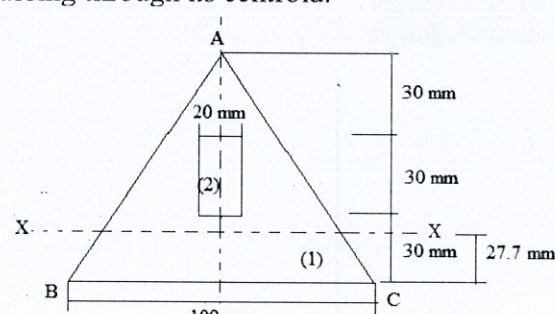
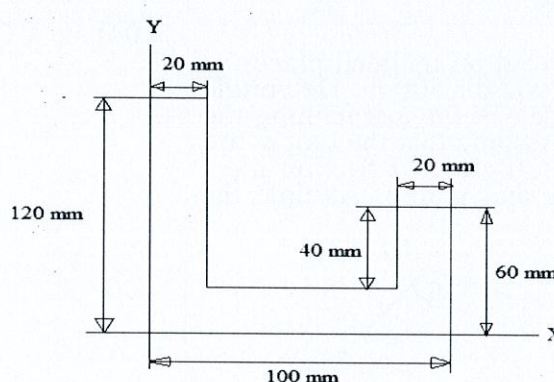
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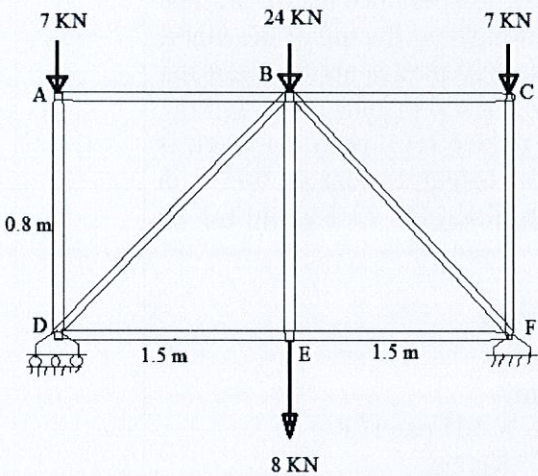
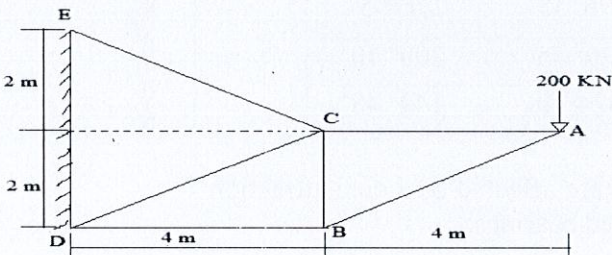
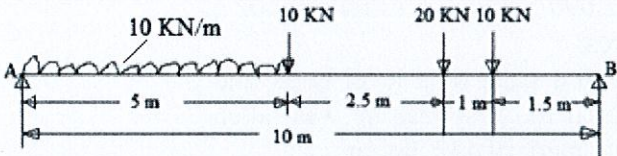
[Max. Marks: 40

TOTAL NUMBER OF QUESTIONS IN THIS PAPER	= 05
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NOTE: Answer all the five questions.

S.No	Questions	Marks	CO	BL	PI
Q.1	(a) What do you mean by system of force? Discuss coplanar concurrent and coplanar non- current system	(02)	CO1	L1	1.3.1
	(b) Define the Polygon law of forces.	(02)	CO1	L2	1.3.1
	(c) The cylinders are piled up in a rectangular channel as shown in fig. (a) Determine the reactions at point 6 between the cylinder A and the vertical wall of the channel. Cylinder A: radius = 4m, weight = 150 N Cylinder B: radius = 6m, weight = 400 N Cylinder C: radius = 4m, weight = 200 N	(04)	CO1	L5	2.2.3
	 Fig. 1				
	OR				
(d)	Two blocks A and B are placed on inclined planes as shown in fig. The block A weight 500 N. Determine minimum weight of the block B for maintaining the equilibrium of the system. Assume that the blocks are connected by an pulley. Coefficient of friction $\mu_A = 0.25$ (between the block A and plane). Assume the same value for $\mu_B = 0.25$.	(04)	CO1	L5	2.2.3
	 Fig.2				

Q.2	(a)	Differentiates between centre of gravity and centroid.	(02)	CO2	L1	2.2.4
	(b)	Find centroid of L- Section as shown in fig.	(02)	CO2	L4	2.2.3
						
	Fig.3					
(c)	A rectangular hole is made in a triangular section as shown in figure. All dimensions are in mm. Determine the moment of inertia of the section about centroidal x-x axis passing through its centroid.		(04)	CO2	L4	2.2.3
		Fig. 4 OR				
(d)	Find the moment of inertia about x-axis.		(04)	CO2	L4	2.2.3
		Fig.5				

Q.3	(a)	Find the forces in the member AB, DB, DE of the truss as shown in fig.by method of section.	(03)	CO3	L5	2.4.1	
							
	Fig.6 OR						
	(b)	Explain calculation of two reactions on a simply supported beam with uniformly distributed load is equal to 10 kN/M and span length is 10 m.	(03)	CO3	L4	2.4.1	
	(c)	Determine the support reactions and nature magnitude of forces in members of truss shown in figure.	(05)	CO3	L5	2.4.1	
							
	Fig.7 OR						
	(d)	Determine the reaction and draw shear force and bending moment for simple supported beam loaded as shown in fig.	(05)	CO3	L5	2.4.1	
							
	Fig.8						
	Q.4	(a)	Explain different types of chain in survey.	(02)	CO4	L1	1.3.1
		(b)	Difference between Prismatic Compass and Surveyor's compass.	(02)	CO4	L2	2.2.4
(c)		A Nominal distance of 3 Metres was set out with a 30 m	(04)	CO4	L5	2.2.3	

	steel tape from a mark on the top of the one peg to a mark on the top of another, the tape being in catenary under a pull of 10 Kg and at a mean temperature of 70⁰ F. The Top of one peg was 0.25 metre below the top of the other. The top of the higher peg was 460 metres above mean sea level. Calculate the exact horizontal distance between the marks on the two pegs and reduce it to mean sea level, if the tape was standardized at a temperature of 60⁰ F in catenary, Under a pull of (a) 8 Kg (b) 12 Kg and (c) 10 Kg. Take radius of earth = 6370 Km Density of Tape = 7.86 g/cm³ Section of Tape = 0.08 sq.cm Coefficient of expansion = 6 X 10⁻⁶ per 1⁰ F Young' s Modulus = 2 X 10⁶ Kg/cm²																			
	OR																			
	(d) The Following bearings were observed while traversing with a compass: <table border="1"><thead><tr><th>Line</th><th>F.B</th><th>B.B</th></tr></thead><tbody><tr><td>AB</td><td>45⁰ 45'</td><td>226⁰ 10'</td></tr><tr><td>BC</td><td>96⁰ 55'</td><td>277⁰ 5'</td></tr><tr><td>CD</td><td>29⁰ 45'</td><td>209⁰ 10'</td></tr><tr><td>DE</td><td>324⁰ 48'</td><td>144⁰ 48'</td></tr></tbody></table> Mention which stations were affected by Local attraction and determine the corrected bearings.	Line	F.B	B.B	AB	45 ⁰ 45'	226 ⁰ 10'	BC	96 ⁰ 55'	277 ⁰ 5'	CD	29 ⁰ 45'	209 ⁰ 10'	DE	324 ⁰ 48'	144 ⁰ 48'	(04)	CO4	L5	2.2.3
Line	F.B	B.B																		
AB	45 ⁰ 45'	226 ⁰ 10'																		
BC	96 ⁰ 55'	277 ⁰ 5'																		
CD	29 ⁰ 45'	209 ⁰ 10'																		
DE	324 ⁰ 48'	144 ⁰ 48'																		
Q.5	(a) What is Levelling? Define Reduced level and Bench mark.	(02)	CO5	L1	1.3.1															
	(b) Describe various methods of contouring.	(02)	CO5	L1	1.3.1															
	(c) The following staff readings were observed successively with a level, the instrument having been moved after third, sixth and eighth readings : 2.228 ; 1.606 ; 0.988 ; 2.090 ; 2.864 ; 1.262 ; 0.602 ; 1.982 ; 1.044 ; 2.684 metres. Enter the above readings in a page of a level book and calculate the R.L. of point if the first reading was taken with a staff held on a bench mark of 432.384 m.	(04)	CO5	L5	2.1.2															
	OR																			
	(d) During a construction work, the bottom of a R.C Chhajja A was taken as a temporary B.M (R.L. 63.120). The following notes were recorded: Reading on inverted staff on B.M, No, A :2.232	(04)	CO5	L5	2.1.2															

	Reading on peg P on ground change of instrument : 1.034 Reading on peg P on ground : 1.328 Reading on inverted staff on bottom of cornice B : 4.124 Enter the readings in a level book page and calculate the R.L, of Cornice B.				
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